

terminal of the charger, that is, with its cathode pointing towards positive terminal of the charger.

The downside of such an arrangement is that during regular current flow, there can be significant power dissipation in the diode. For example, at 20A and 0.7V drop across the diode, the diode dissipates 14 watts, while the diode also reduces the charging voltage by its forward drop value.

A slightly better way is to use a MOSFET switch, and drive it based on a signal that ensures that the battery voltage is always lower than the charger voltage. The lower channel resistance of a MOSFET, called R_{DS-ON} , results in reduced power dissipation as compared to the diode example. For a typical R_{DS-ON} value of 20 mOhm, at a charging current of 20A, power dissipation would be about 8 watts.

Another scheme can use a normally-open contact of an electromagnetic relay and drive the relay with the same above signal. This allows almost zero power dissipation across the switching contacts. Both these types of switches need to have higher voltage ratings than the main charging switches.

Deep-discharge protection

This is an often overlooked yet extremely important protection mechanism for the battery itself. This arrangement is typically part of a charge-controller and/or battery management system. It paves the way for enhancing the battery's life significantly. Once a battery discharges to a very low voltage, such that its depth of discharge reaches approximately 80% of its fully charged capacity, any further discharge may turn out to be fatal for the battery.

This applies to all battery chemistries, and this fatal electrochemical effect occurs in different batteries via different mechanisms, but the final effect is similar. Except that

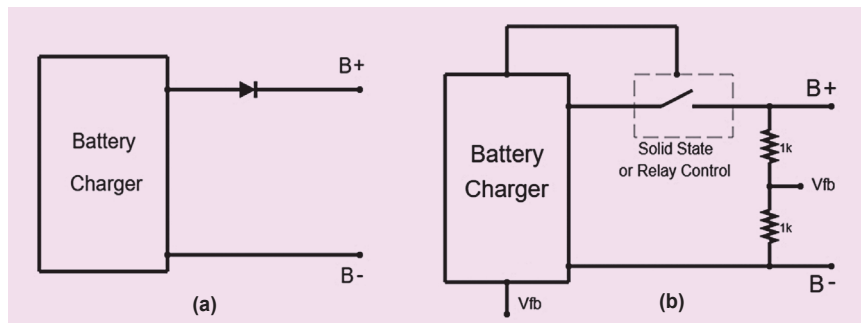


Fig. 4: (a) Forward biased diode at the output of the charger, (b) MOSFET switch or relay contact at output of the charger

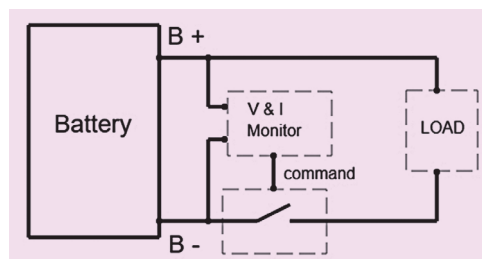


Fig. 5: Circuit to monitor battery discharge and disconnect battery from load

in the energy-dense chemistries, such as Li-ion, the deep discharge can result in swelling and thermal runaway of the battery. Many informative articles about the deep discharge phenomenon in Li-ion batteries can be found in journals such as *Nature*.

In lead-acid batteries, deep discharge can lead to 'shedding' of the positive active material and shorting of the plates. So, in all cases, deep discharge of batteries is best avoided. The protection here is slightly different, and more robust from idle state mechanism, where the controllers are placed in an idle condition using their sleep modes to save battery current draw on an ongoing basis. This protection is used when the battery has actually been depleted extensively.

This protection is implemented using a circuit that continuously monitors the battery terminal voltage and battery current draw while it is being discharged, thereby estimating its depth of discharge (DoD) or state of charge (SoC). Once the SoC goes below a certain pre-decid-

ed value, the circuit drives a switch that disconnects the battery from the load.

While the main power load is disconnected, trickle current consumed by the control circuitry may still keep the battery discharge going. Though this trickle current may be small, at about 500 to 1000 times lesser than overall cur-

rent draw, but it is continuous. And this continuous draw can pull the battery into the fatal zone. Hence, a further enhancement to the scheme ensures that either the control circuitry is also disconnected (good for pure analogue controllers), or the digital controller itself is placed in a sleep condition.

Controllers should be chosen with as low sleep currents as possible. Also, once the battery is disconnected, the electrochemical nature of the battery allows its terminal volage to build up again to a value where the controller might see it as healthy once again. This should not allow the battery to be reconnected to the load, otherwise an on/off sequence would occur, which is detrimental to the battery system and anything that draws power from it.

A separate circuit allows this deep-discharge shutdown to be reset and the battery system to reconnect to the load. However, it is best to interlock this with an input charging supply, such that the reset occurs automatically once charging starts. **EFY**